



Ministry of Foreign Affairs

Veilige communicatie in het quantum tijdperk

Microsoft Security NL Meetup

11-12-2025



Anco Stuij

PQC & QKD Security Architect **BZ Quantum Veilig**



Ministry of Foreign Affairs

Rubicon



Agenda

1. Ministerie van Buitenlandse Zaken
2. Quantum 101
3. PQC & QKD
4. QCINed (Quantum Computing Infrastructure Nederland)
5. Demo
6. Microsoft's Quantum Safe Program (QSP)
7. Call to Action

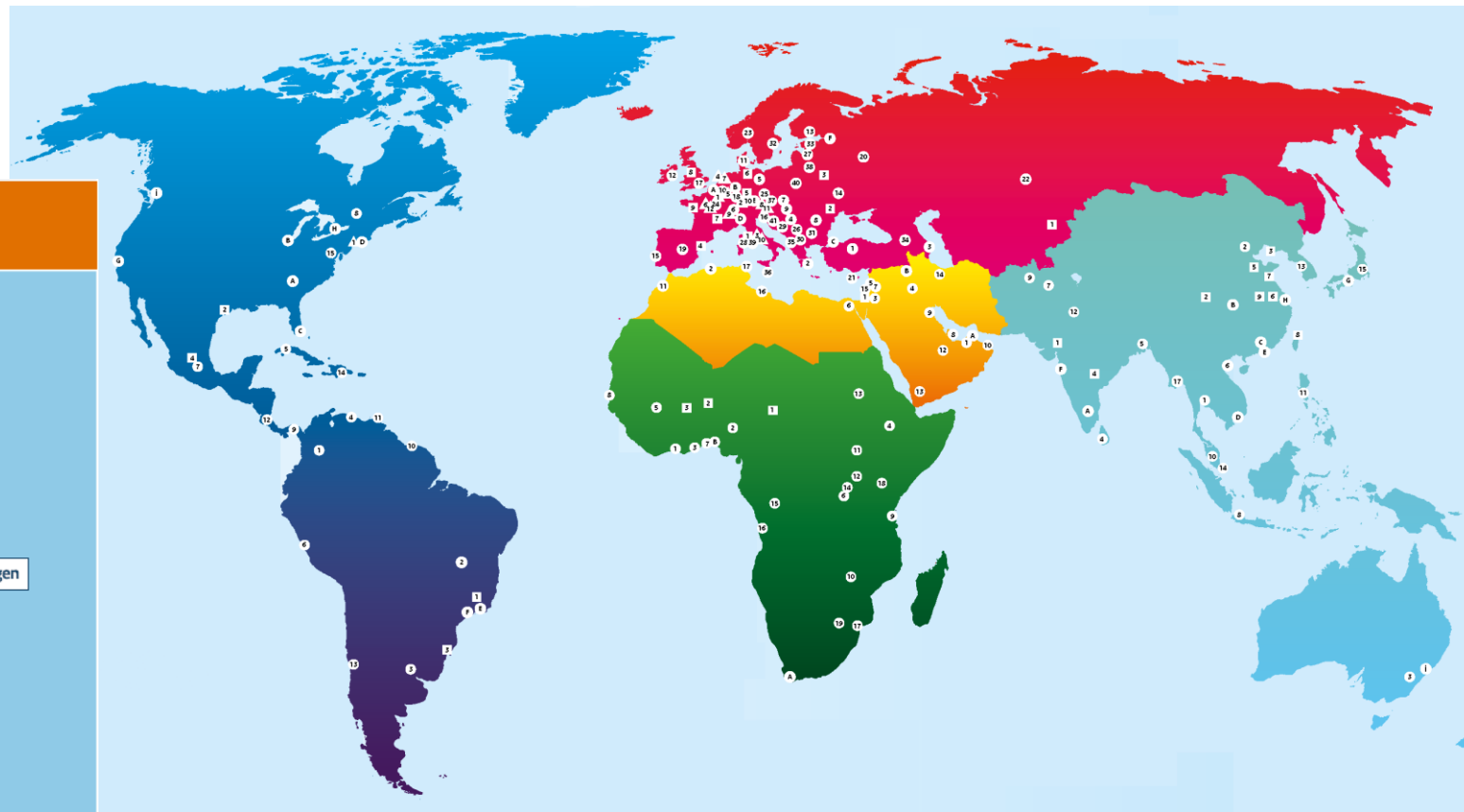
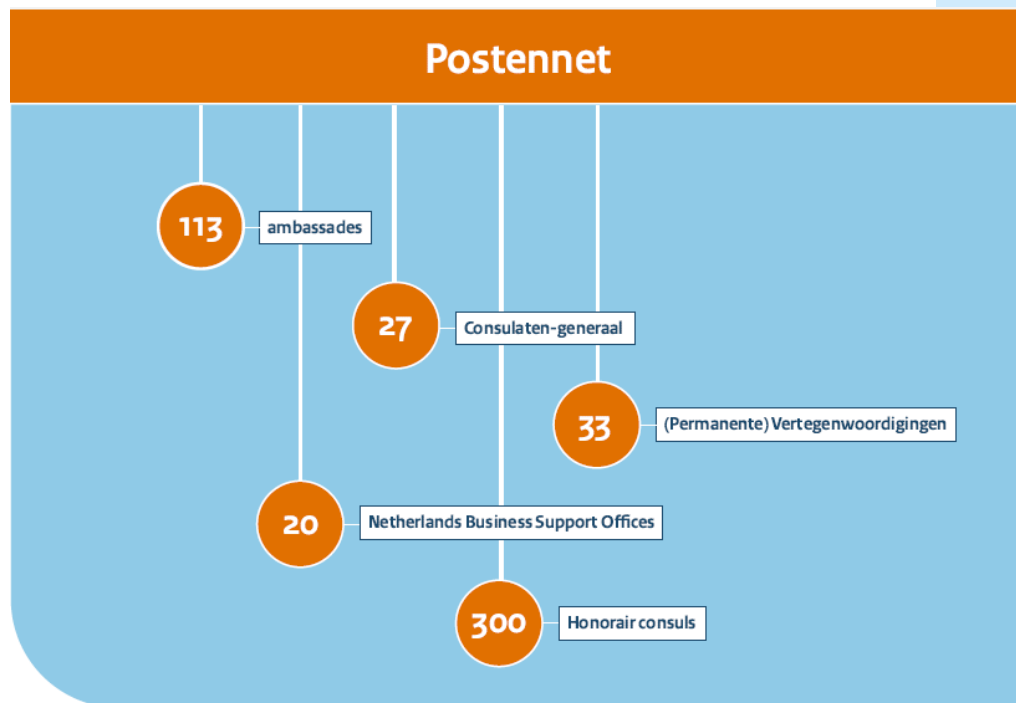


We zijn er. Voor Nederland, wereldwijd.

Het Ministerie van Buitenlandse Zaken coördineert en voert het buitenlands beleid van de Nederlandse regering uit. Het behartigt Nederlandse belangen in het buitenland, ondersteunt Nederlanders die in het buitenland wonen of reizen, en vertegenwoordigt Nederland bij internationale organisaties. Daarnaast werkt het ministerie aan vrede, veiligheid, duurzame ontwikkeling en internationale samenwerking op thema's als klimaat, mensenrechten en ontwikkeling.



Het Postennet



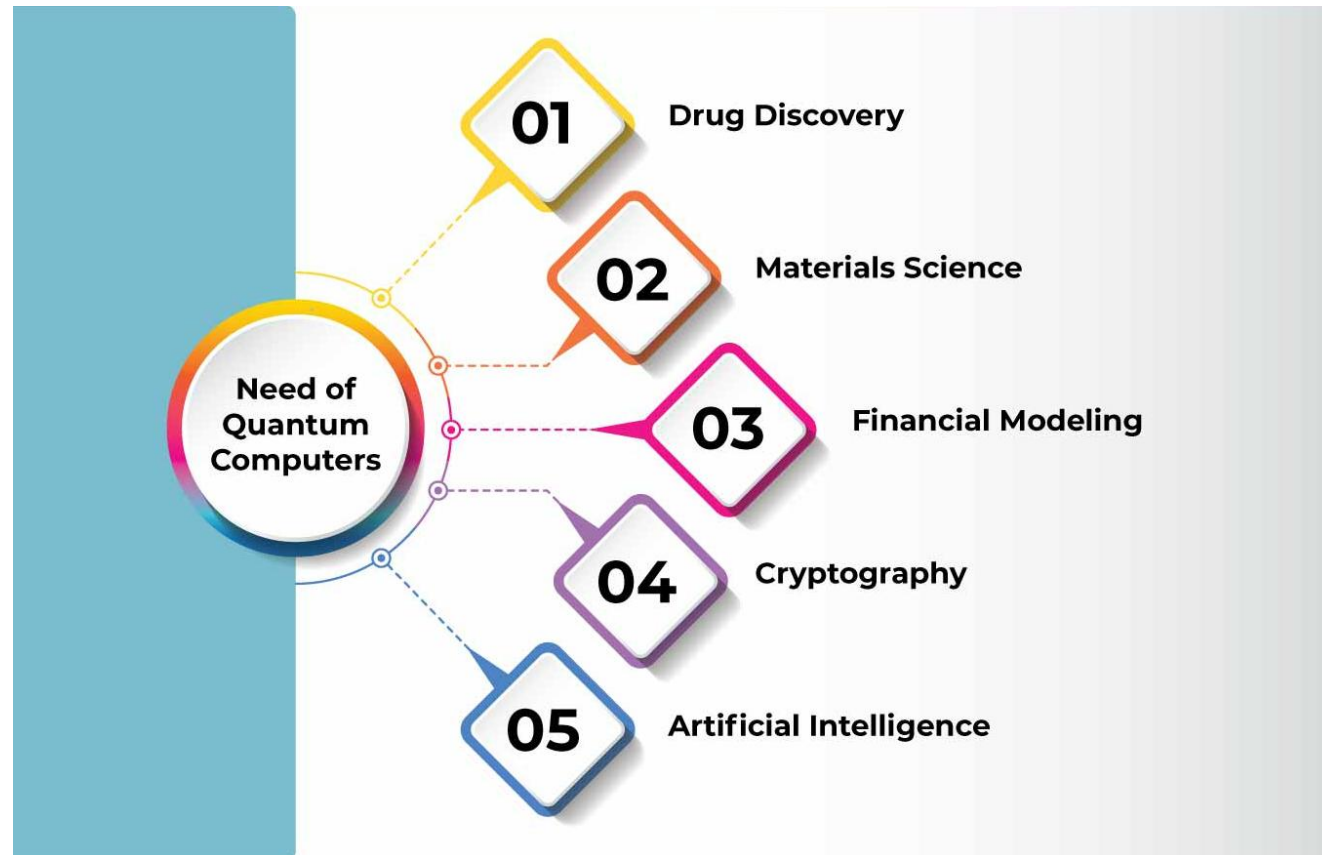


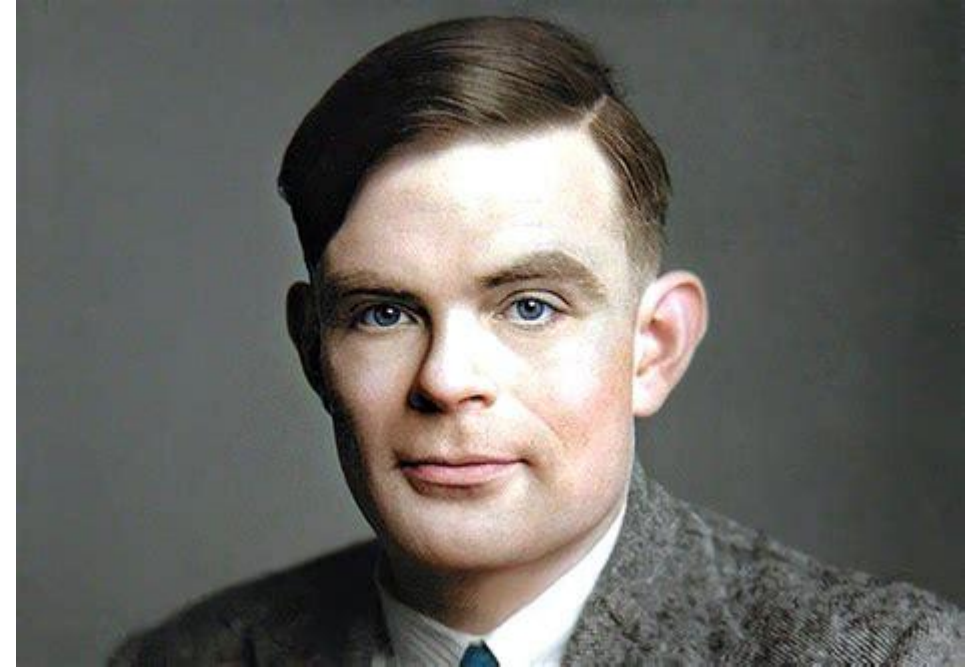
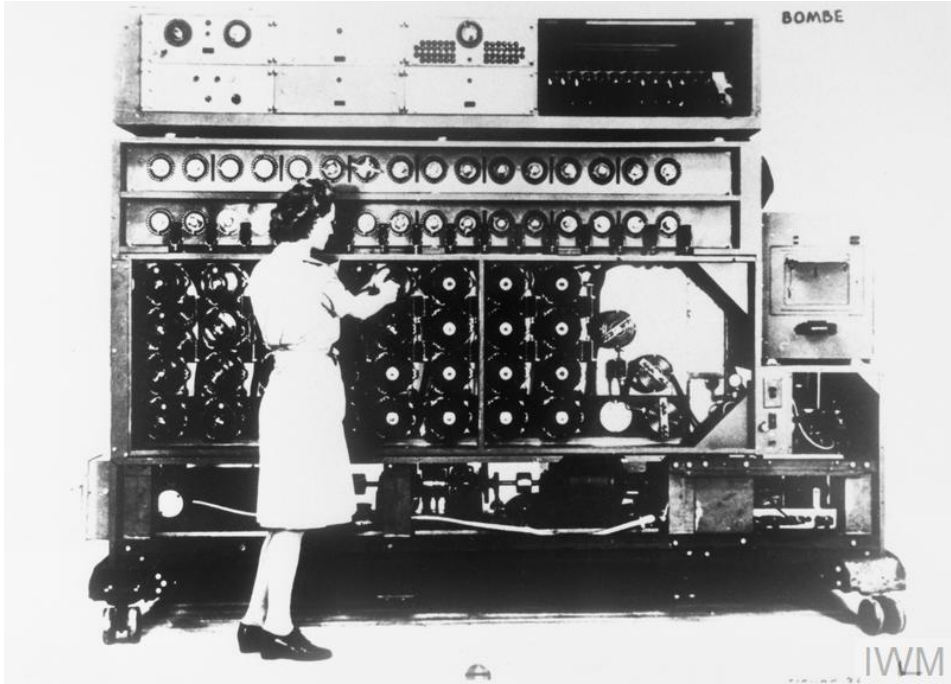
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Quantum for Good

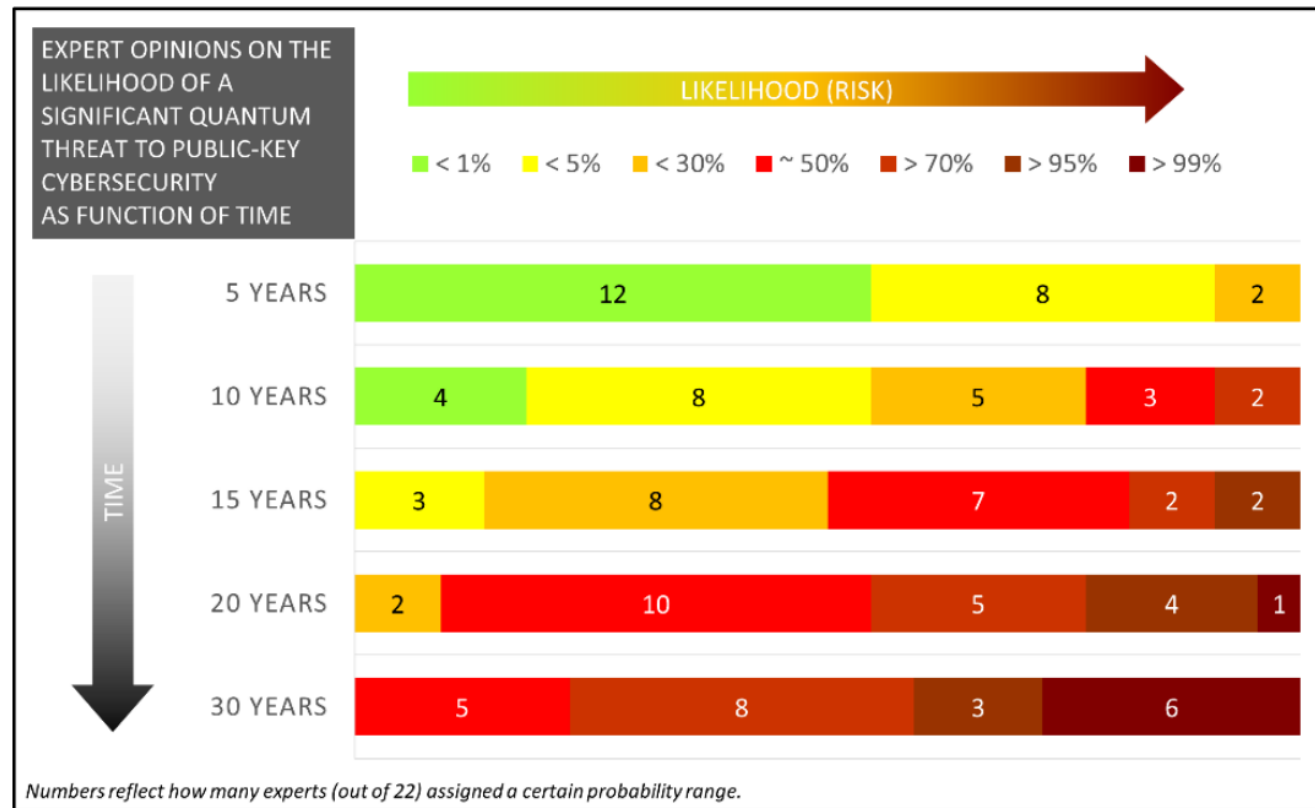




The second world war,
was won by the allies in Europe,
by breaking cryptography with a machine.



Are we there yet?



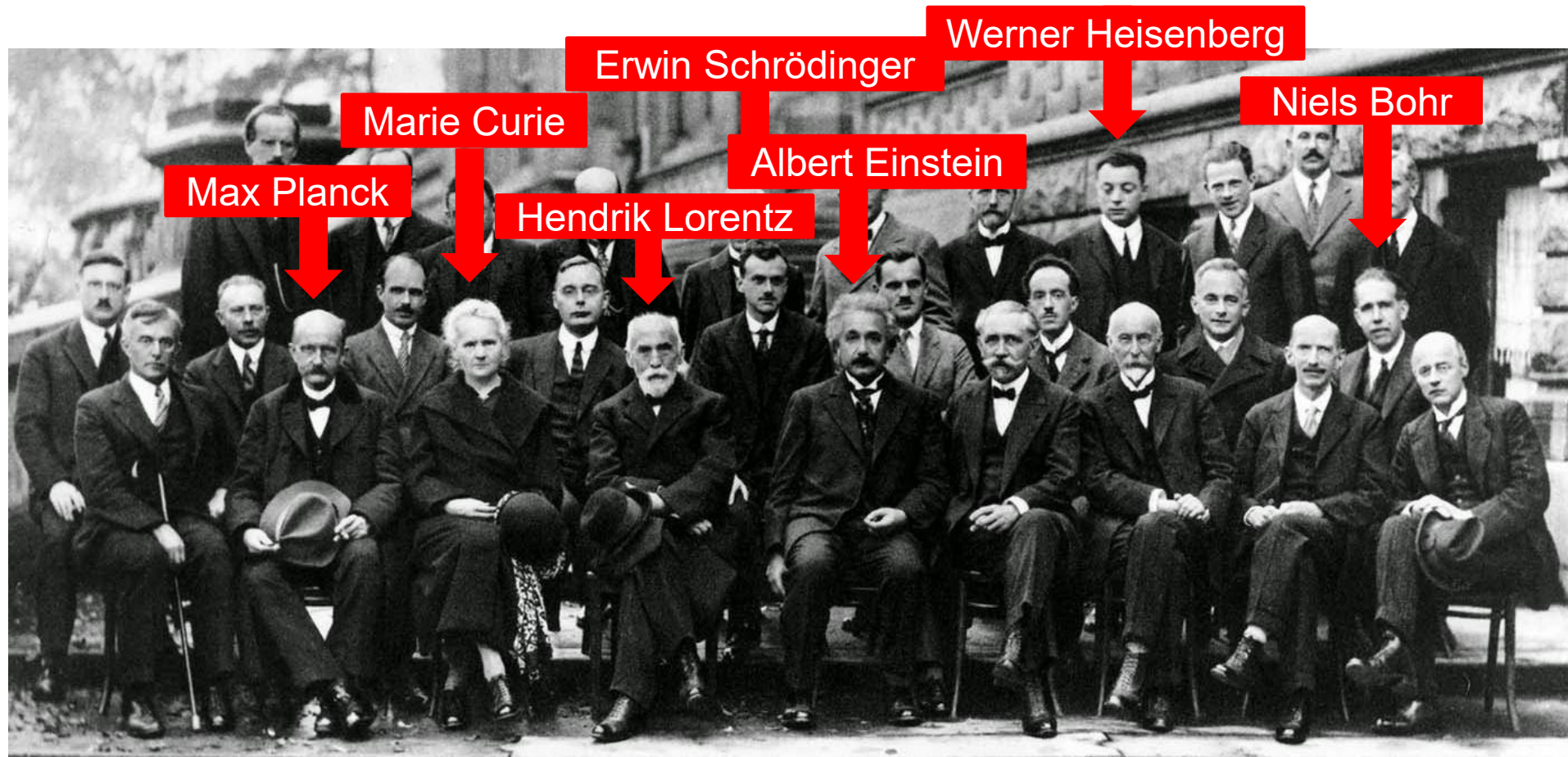


Quantum History

- 19th century
 - 1805 Thomas Young double-slit experiment
- 20th century
 - 1900 Max Planck introduces the quantum theory
 - 1913 Niels Bohr presents the atomic model
 - 1926 Werner Heisenberg and Erwin Schrödinger formalise quantum mechanics
 - 1935 Einstein, Podolsky and Rosen describe the EPR-paradox
 - 1936 Alan Turing presents the Turingmachine
 - 1949 Richard Feynman ontwikkelt zijn padintegraalformulering
 - 1980 Paul Benioff beschrijft het eerste quantummechanisch model
 - 1980 Yuri Manin describes the idea of quantum computing
 - 1994 Peter Shor publishes a quantum algorithm for factorisation
 - 1995 Lov Grover publishes a quantum search algorithm
- 21st century
 - 2019 Google claims quantum supremacy with 53 qubits
 - 2019 IBM launches the first quantum computer for commercial use
 - 2025 Google demonstrates the Willow-chip with 105 qubits
 - 2025 Craig Gidney shows that RSA-2048 can be broken using less than 1 million qubits

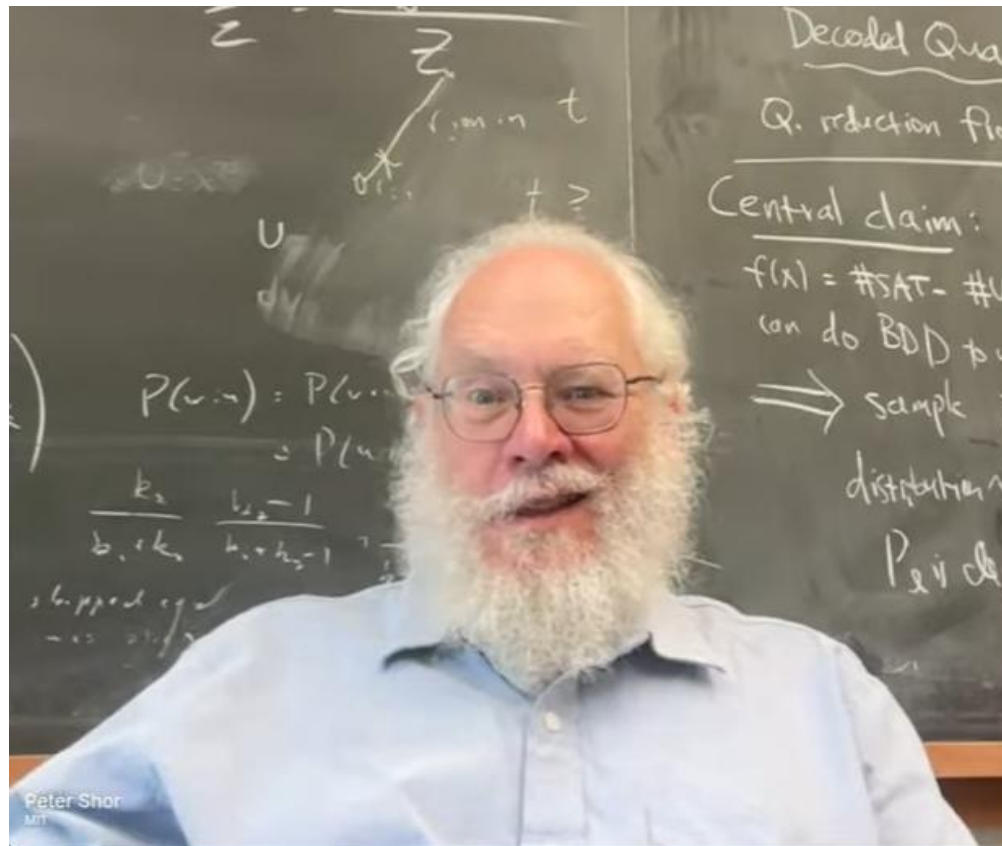


Quantum History - 1927 Solvay Conference





Quantum History – 1994 Shor's algorithm





Quantum Technology

1. Quantum Computing

Powerful calculations with qubits, which can assume multiple states at the same time (superposition) and influence each other (entanglement), allowing complex calculations to be performed much faster than with classical computers.

2. Quantum Communication (QKD)

Extremely secure data transfer via Quantum Key Distribution (QKD), detecting any eavesdropping attempt by measuring quantum states interfering with key transfer.

3. Quantum Sensing

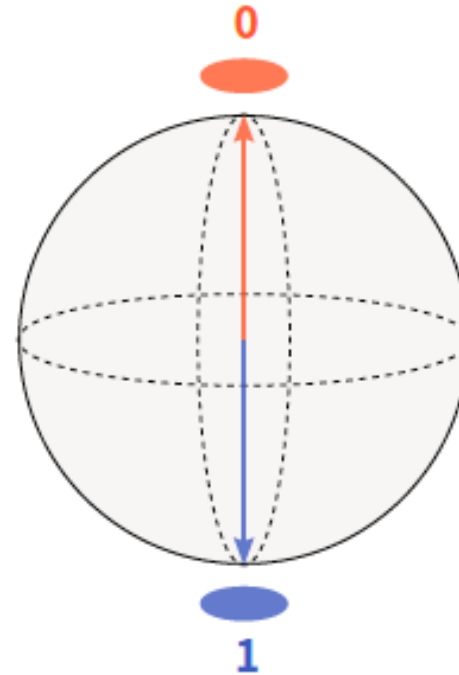
Detecting extremely small changes in, for example, magnetic fields or gravity, with applications in security and infrastructure.



Concepts in Quantum Mechanics

1. Superposition
2. Measurement
3. Entanglement

Q-Bit





1. Quantum Superposition



Superposition is the ability of a quantum system to be in multiple states at the same time until it is measured.

Thought experiment:
Spin a coin, while spinning it appears to be neither heads nor tail.



2. Quantum Measurement



Measurement is the process that turns a quantum state in a definite outcome.

Thought experiment:

Spin a coin, while spinning it appears to be neither heads nor tail. When the spinning stops the state is measured and the superposition is broken.



3. Quantum Entanglement



The bits are **entangled** and “the value of each bit is interconnected” within a system.

Thought experiment:
Imagine, you find one sock, and don't know where the other is...

Choose which side will you put the sock on?



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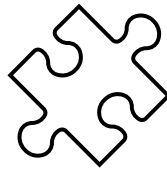
PQC & QKD

1. **Post Quantum Cryptography (PQC)**

Cryptographic algorithms that are resistant to attacks from the crypto-relevant quantum computer (CRQC) based on mathematical problems.

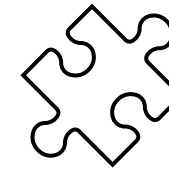
2. **Quantum Key Distribution (QKD)**

Extremely secure data transfer via Quantum Key Distribution (QKD), detecting any eavesdropping attempt by measuring quantum states interfering with key transfer.



Quantum internet &
Quantum Key Distribution (QKD)

Quantum physics
Information theoretical safe
Encryption Key Distribution

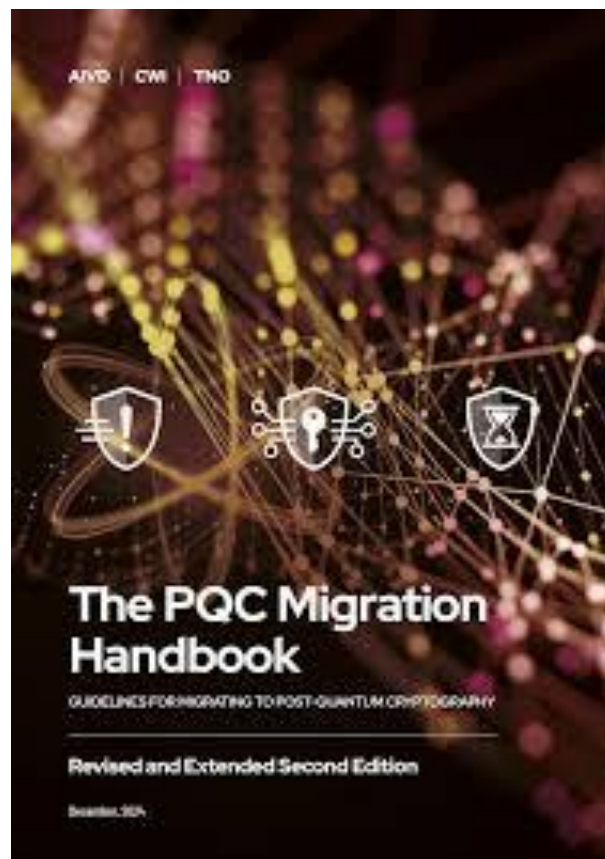


Classic Technology
Post Quantum Cryptography (PQC)

Mathematics
Information theoretical unsafe
Computational safety
Doesn't cover all kinds of attacks
Multi functional



AIVD - The PQC Migration Handbook





AIVD Position Paper on QKD



Federal Office
for Information Security



General Intelligence and
Security Service
*Ministry of the Interior and
Kingdom Relations*

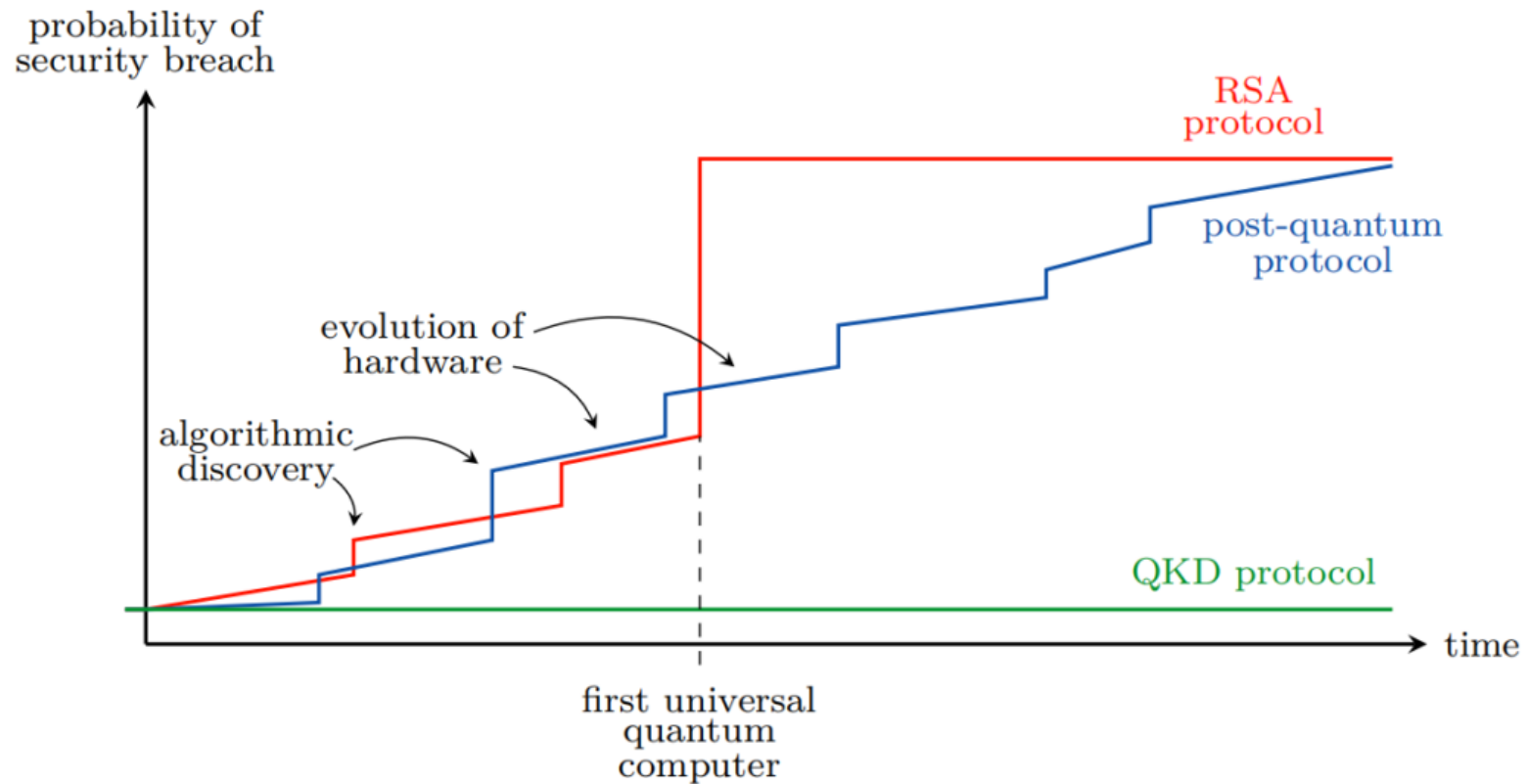


SWEDISH ARMED FORCES

Position Paper on Quantum Key Distribution



Rebuttal Paper Renner & Wolf





EU Roadmap Post Quantum Cryptography (PQC)

1. 31-12-2026
 - Crypto Asset Management (CBOM)
 - Create a national awareness program
 - Share knowledge
 - Develop a timeline
2. 31-12-2030

High risk systems are Post Quantum Ready
3. 31-12-2035

All systems are Post Quantum Ready



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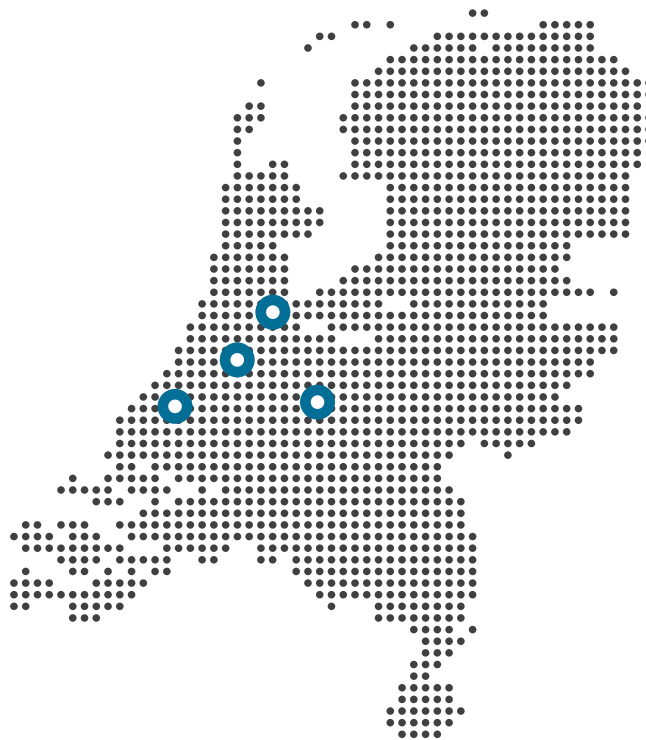
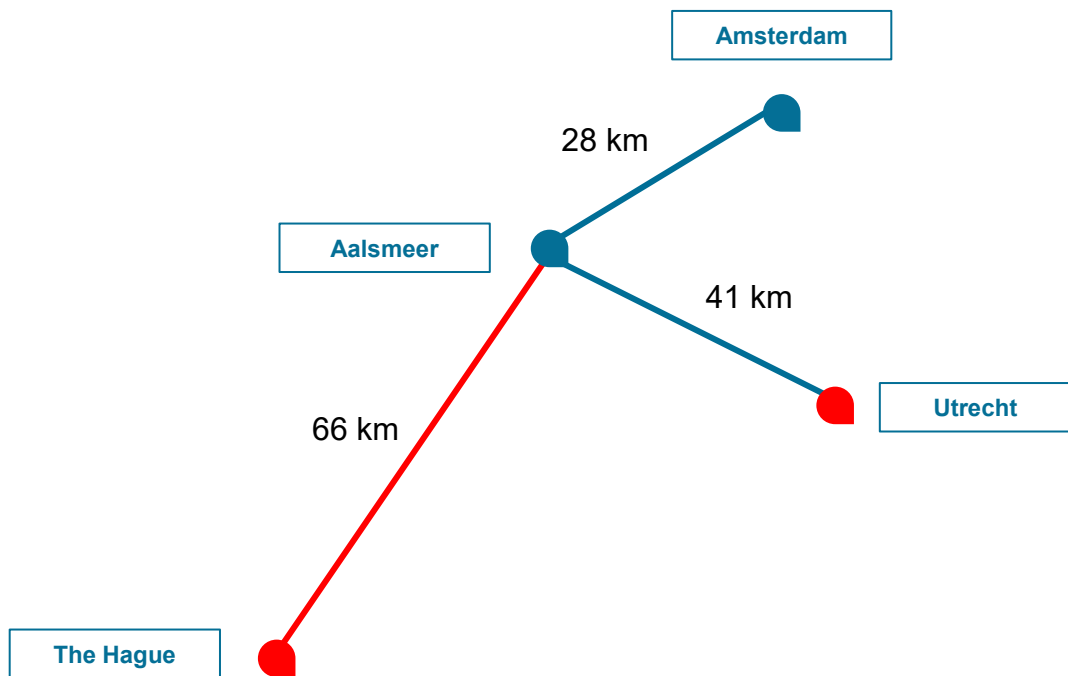
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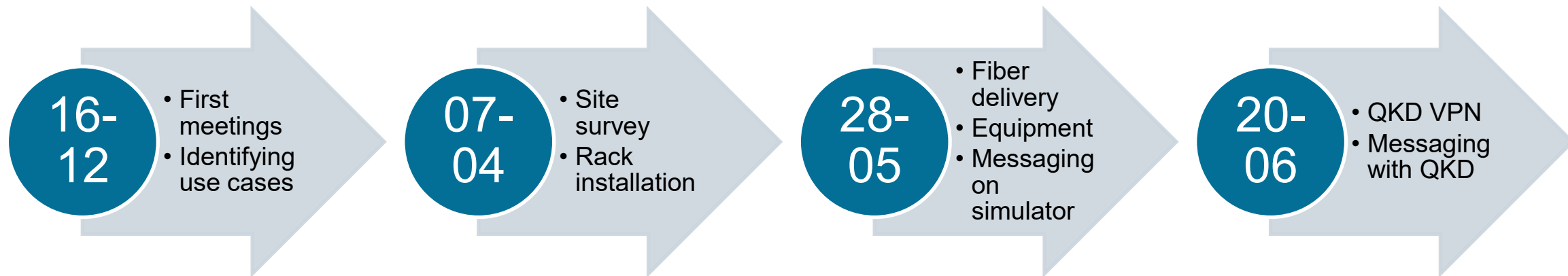


QKD infrastructure (QCINed)





Timeline





07-04 Rack installation





27-05 Fiber delivery



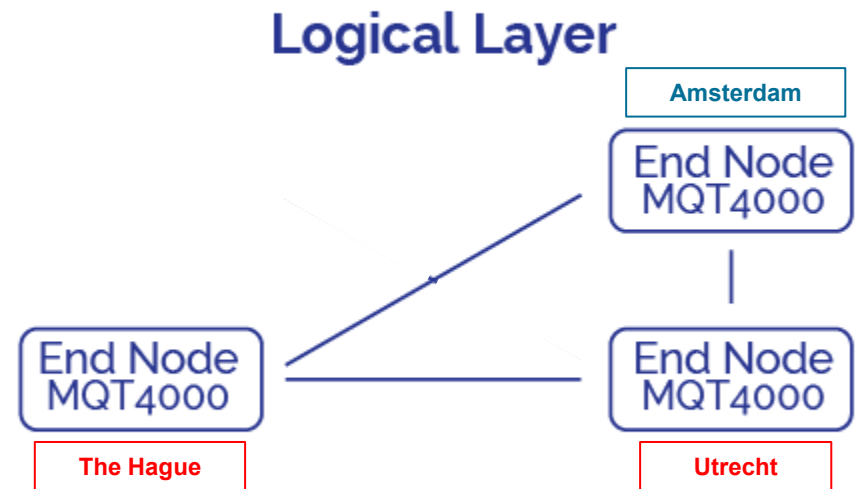
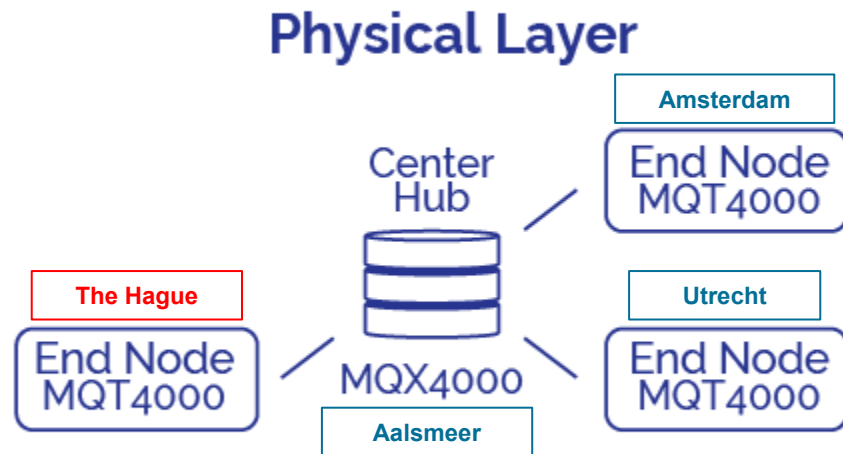


28-05 Racking the equipment



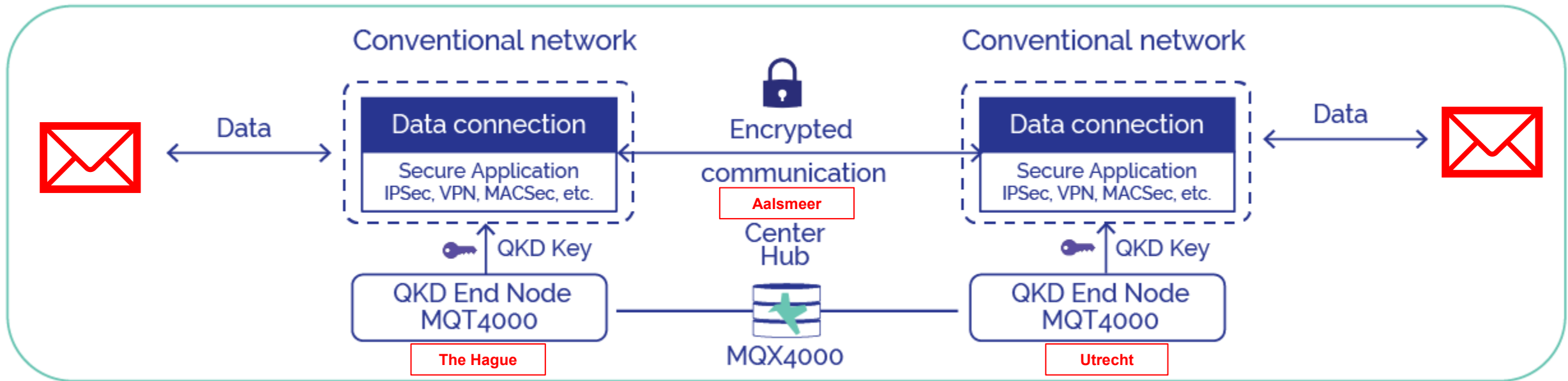


QKD link infrastructure 1/2





QKD link infrastructure 2/2





Future QCINed

1. Uitbreiden naar België, Luxemburg en Duitsland over land
2. Uitbreiden naar Bulgarije en Cyprus via satelliet



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QKD Use Cases

Use case 1

QKD VPN

Secure a VPN with QKD

Use case 2

QKD Messaging App

Secure an application with QKD



Use case 1: QKD VPN

- VPN secured with QKD
- Network layer
- IPsec protocol
- Secures all communication



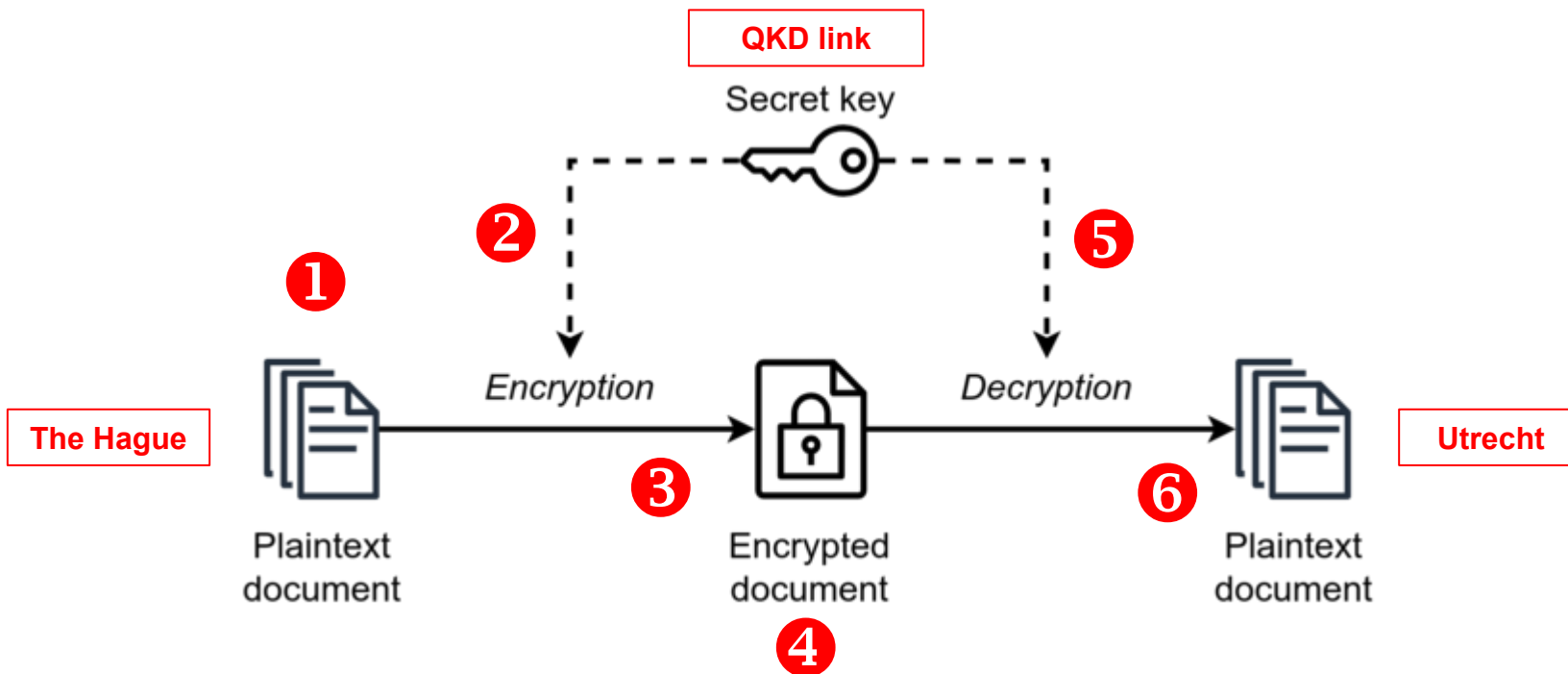


Use case 2: Diplomatic Cables application

- Provides formal communication between the departments in The Hague and the Dutch missions abroad.
- Contains:
 - Diplomatic cables (e.g. minutes of international meetings or situation reports)
 - Other messages with a “formal” status such as minutes of board meetings or special announcements, etc.
- Messages are assigned a classification.
- Application is web based and includes text editor, database, roles/authorizations and email integration.



QKD Messaging App demo







PQC loves QKD

QKD VPN

Combines PQC algorithm ML-KEM768 with the ETSI GS QKD standard.

QKD Messaging App

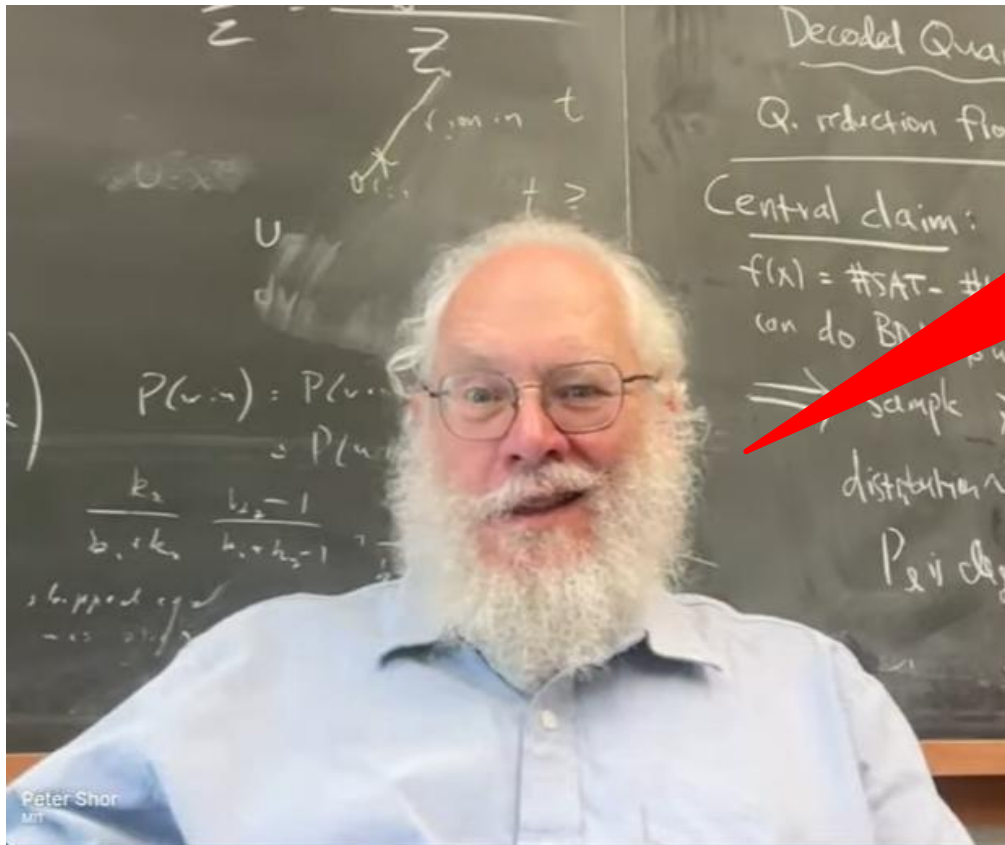
Combines the ETSI GS KQD standard with the AES 256 algorithm.

Combine the power of math with the law of physics to secure your communication in the quantum era.





Peter Shor



“My algorithm makes QKD necessary.”

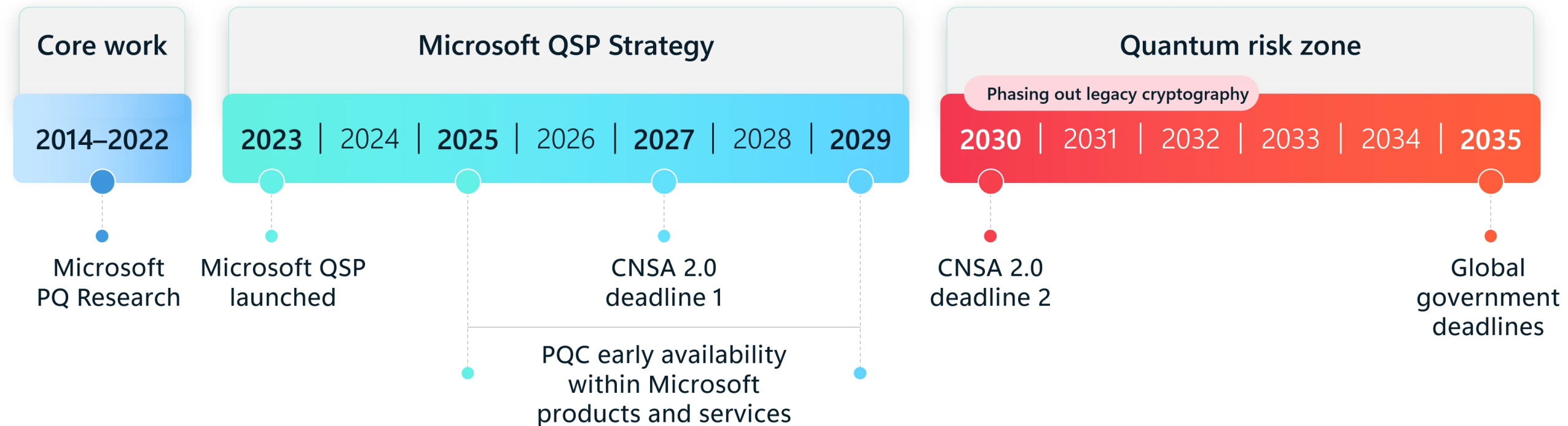


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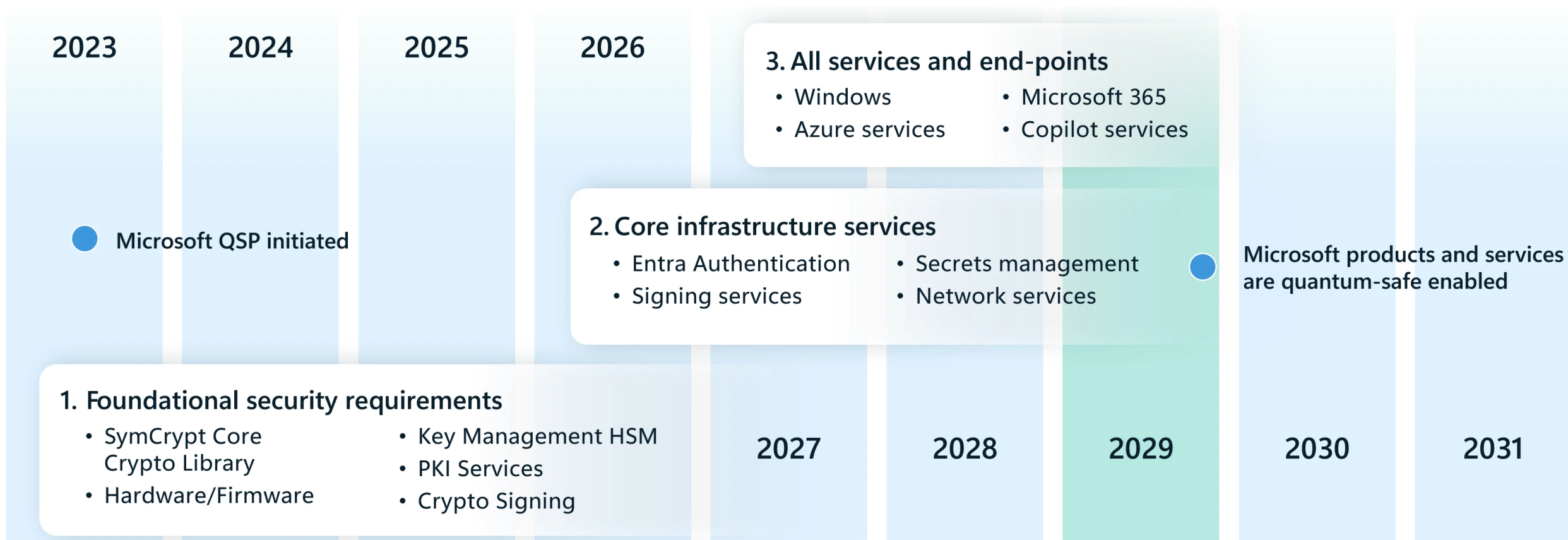


Microsoft Quantum Safe Program





Microsoft QSP timeline





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Call to Action

1. Start to assess your current cryptographic inventory (CBOM)
2. Run [CodeQL](#) to scan your codebase for PQ vulnerabilities
3. Upgrade .NET applications to .NET 10 (SymCrypt library)
4. Update mobile OS to iOS 26+ and Android 17+ (unannounced)
5. Update Windows clients and servers to November 2025 builds (SymCrypt library)
6. Linux and Unix distros starting to support PQC like Debian 13, Ubuntu 25.10 etc.
7. Start testing enforcement of TLS 1.3 in your Azure workloads
8. Experiment with PQC-ready certificates and automatic renewal with ACME protocol



Become Crypto Agile!

